অলীকবচন

Pseudo-Context Detection and Neuro-Symbolic Routing for Bengali LLM Hallucination Detection

IUT 12th ICT Fest Datathon 2026 · Phase-2 In-Person Final

0.937

F1 on the hallucinated class
fully offline · zero-diff rerun

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1 2 3 4 5 6 7 8 9

nine deterministic stages · one neural pass

The task

context reference text — sometimes

prompt_bn a Bengali question

response_bn an LLM's answer

0 / 1

hallucinated / faithful

2,516

Phase-1 test rows; 5,000-row hidden fold in Phase 2

F1o

scored on the hallucinated class only

1.23×

a recovered miss is worth ~1.23× a corrected false alarm — the decision threshold must lean sensitive

Binary faithfulness judgment over (context, question, answer) triples.

The discovery: 17% of the test set lies about itself

prompt

'ঐচ্ছিক' শব্দের বিপরীত শব্দ কী?

context

বাংলা ব্যাকরণ অনুযায়ী 'ঐচ্ছিক' শব্দটির একটি সুনির্দিষ্ট বিপরীত শব্দ রয়েছে।

"According to Bengali grammar, the word has a definite antonym." — asserts an answer exists, provides nothing to check against.

≈430

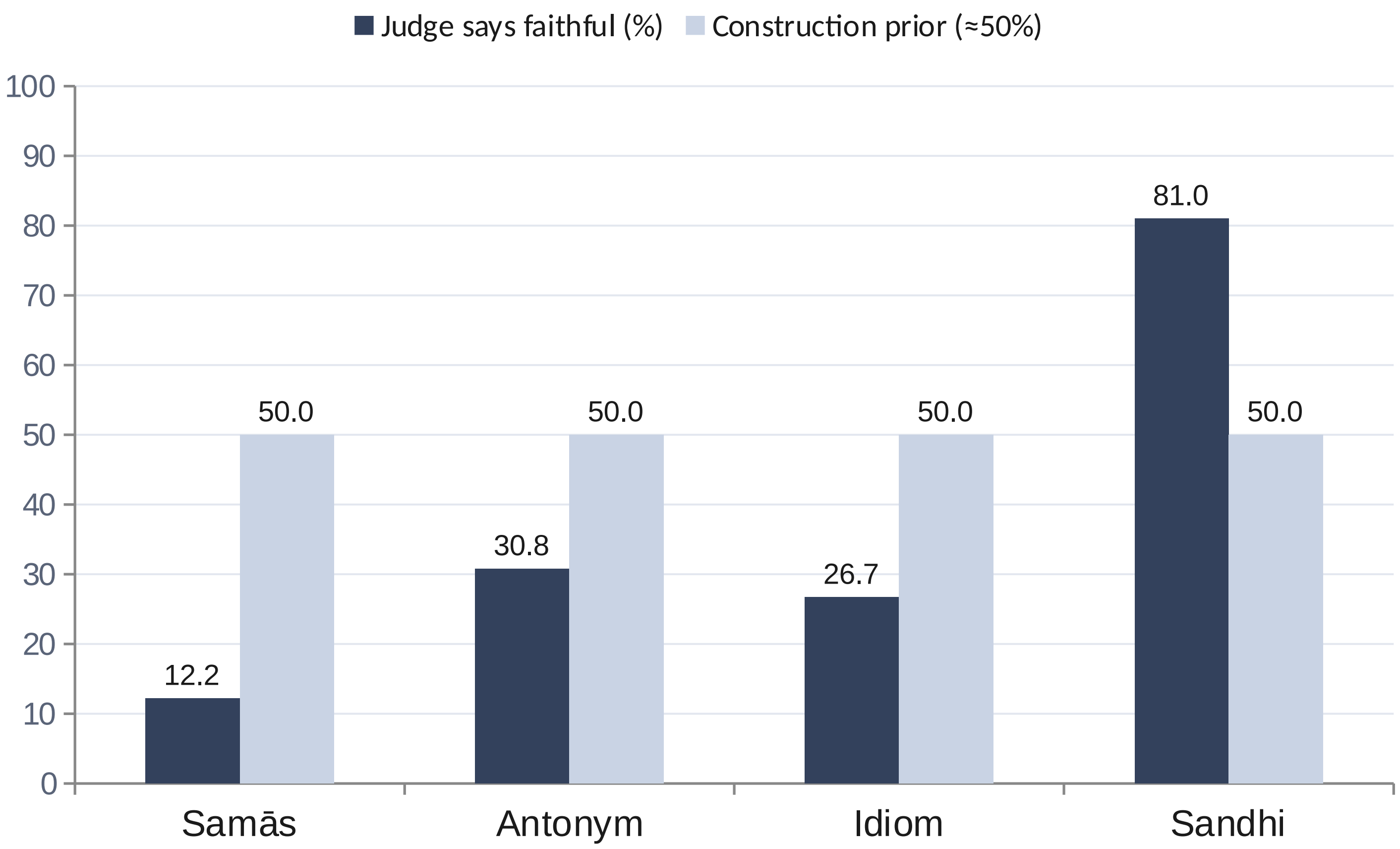
pseudo-context rows (≈17% of test)

On these rows the gold label is **closed-book factual correctness**, not context-groundedness.

Families: সমাস · সন্ধি · বিপরীত শব্দ · বাগধারা · উপসর্গ

A reference-grounded verifier answers a different question than the benchmark asks — and inverts.

Proof: a grounded judge inverts on these families

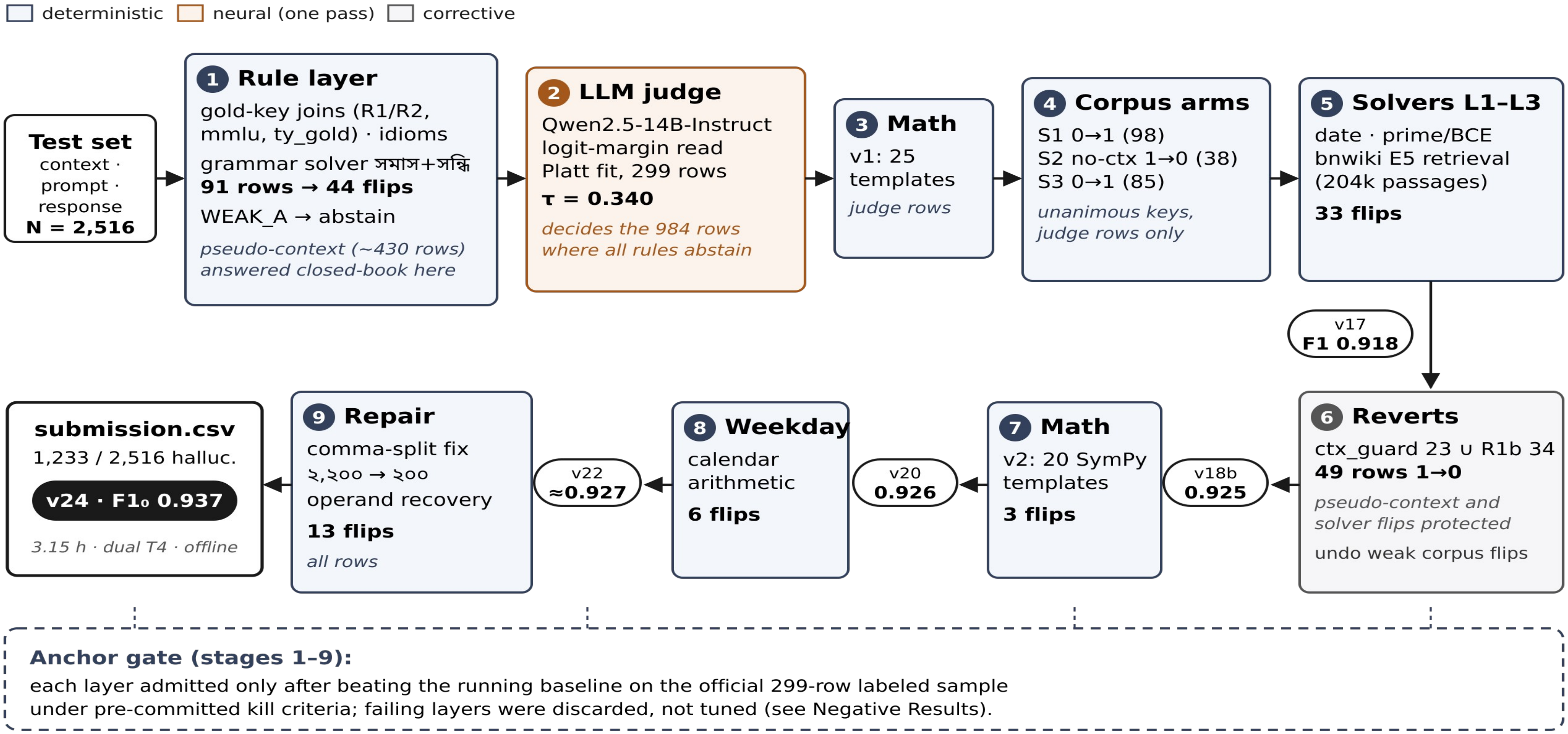


সমাস at 12% faithful is impossible for a balanced generator.

Deriving the answer from each ব্যাসবাক্য symbolically gives 25/49 correct — the 50/50 split the generator intended. The judge is answering “does the context entail this?”; the benchmark is asking “is this correct?”

Sandhi stays high because the surface form itself carries the cue.

Nine deterministic stages around one neural pass



The symbolic layer: prove it or abstain

Grammar solver (সমাস + সন্ধি)

18 token-based rules derive the answer from the context itself — fires on 91 rows, 44 flips, base-independent. Finding: regex \b word boundaries silently fail on Bengali combining marks.

Gold-key joins

Normalized exact joins against declared QA pools (BenHalluEval-derived, TyDiQA-Bn, bangla-mmlu). Match gold → faithful; match a known distractor → hallucinated; ambiguous families abstain by construction.

Computational verifiers

45 math templates (regex + SymPy), weekday arithmetic, date / prime / BCE checks, E5 retrieval over 204k Bengali Wikipedia passages — compute the truth where the truth is computable.

Every rule returns a verdict only inside verified competence — otherwise it abstains and defers to the judge.

The one neural pass: a calibrated logit-margin judge

Qwen2.5-14B-Instruct

- Reads the logit margin between verdict tokens — no generation, no sampling, fully deterministic (~2 s/row)
- Platt scaling fit live on the official 299-row anchor, inside the notebook
- Threshold τ = 0.340, chosen under the $F1_0$ objective — deliberately on the sensitive side
- Decides the 984 rows on which every rule abstains

0.762

judge-only $F1_0$ on the anchor — the floor everything else builds on

0

sampled tokens anywhere in the pipeline — determinism is why the rerun reproduced zero-diff

Data forensics: the inputs were corrupted too

as shipped

২,২০০ → ২০০

Bengali comma-separated numerals truncated at the comma in questions 183–241

repaired deterministically

bisection over candidate restorations — accept only if exactly one makes the word problem internally consistent, then re-solve

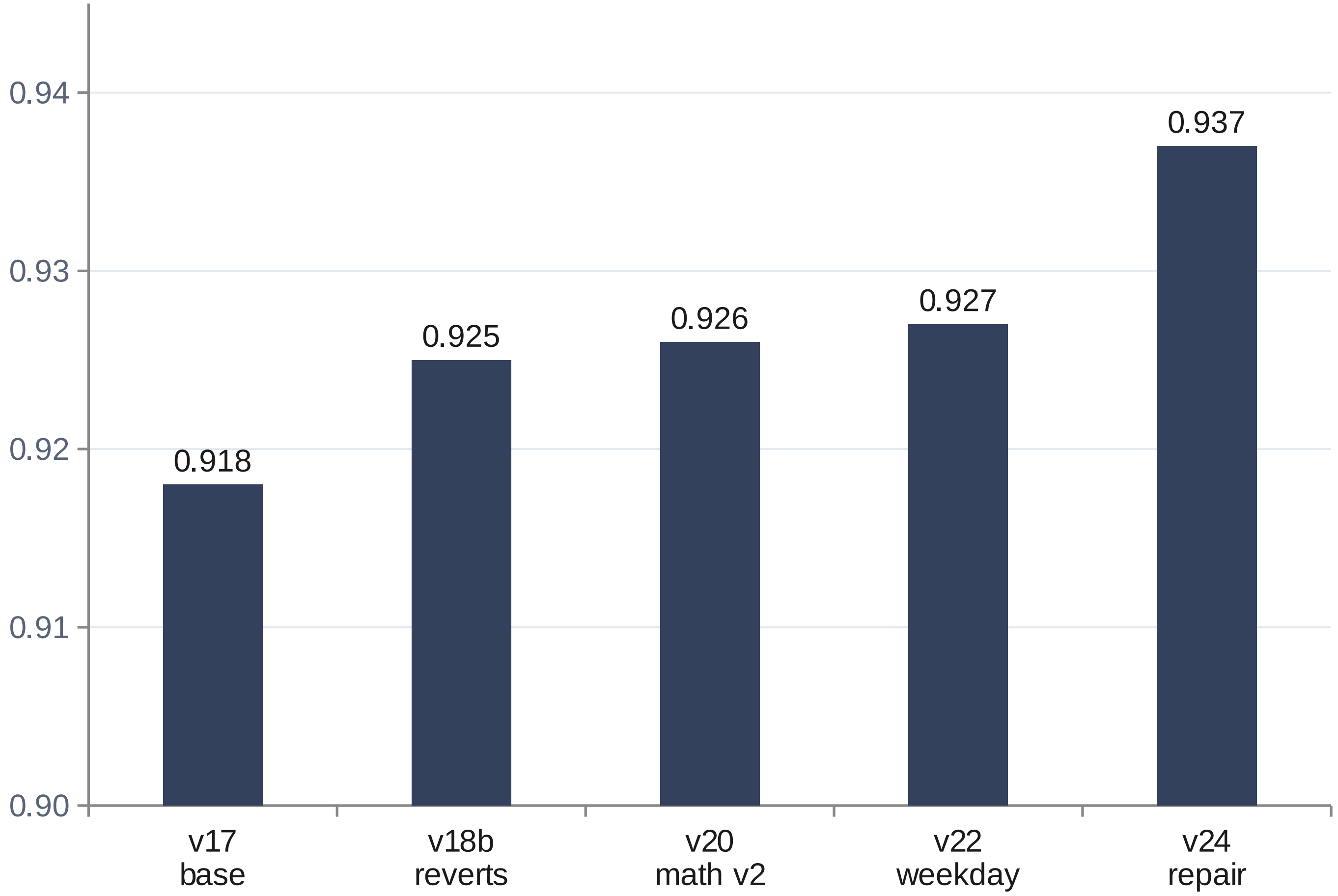
+0.009

the largest single step after the base system (13 flips)

On a benchmark about corrupted outputs, the decisive residual errors were corrupted inputs.

A pure function of raw test text — no row IDs, no manual labels. Fully rule-compliant and fold-portable.

Results, gated



Anchor gate

every layer must beat the running baseline on the official 299-row sample

Pre-committed kill criteria

decided before building; honored regardless of sunk cost

Zero-diff rerun

3.15 h, dual T4, internet off — the organizers' two-run protocol confirmed it

What failed — the gates are the method

32B judge

+0.003 but unpackageable for the rerun — rejected on reproducibility

Disagreement veto

anchor collapsed 0.888 → 0.828 — killed by its pre-committed criterion despite targeting a real defect

Prompt ensemble

−0.007 on the leaderboard

QLoRA fine-tuning

failed on three environment configurations

Generative few-shot judging (AWQ)

$F1_0$ 0.717 vs 0.814, 13.2 h > budget — sampling added noise, not signal

Routing beyond solver-covered families

refuted by anchor evidence — the phenomenon is real, but exploiting it requires deriving the answer

Reporting these is not humility theater — pre-committed gates are what kept negative-value layers out of the shipped system.

Two takeaways

1. Mixed benchmarks should document which task frame each row belongs to.
2. Detection systems should treat identifying the frame as part of the task — then use the cheapest sufficient tool, with calibrated abstention.

0.937 F1o

offline · deterministic · reproduced zero-diff

ধন্যবাদ — **Thank you**

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Backup: pseudo-context families in detail

Family	n	Judge faithful	Mechanism used
সমাস (compound)	49	12.2%	18-rule token classifier on the ব্যাসবাক্য — 25/49 correct (51%)
সন্ধি (euphonic)	42	81.0%	surface form ≠ naive concat unless 'অপরিবর্তিত'
বিপরীত (antonym)	52	30.8%	gold-key / curated pools where available; judge otherwise
বাগধারা (idiom)	30	26.7%	figurative-meaning pool join (idiom arm)

Judge rates from the pre-routing diagnostic snapshot; construction prior ≈50% per family.

Backup: reproducibility & compliance

Two-run protocol

Run 1 reproduces Phase-1 predictions from the raw test file (zero differing rows); run 2 scores the hidden 5,000-row fold. Both completed.

Determinism by construction

Logit-margin read (no sampling) + Platt fit live on the anchor + pure-function rule stages. Checkpoint/resume on judge margins survives session interruptions.

Runtime & footprint

3.15 h on Phase-1 (2×T4, internet off); ≈3.4 h projected on the fold. Model weights well under the 50 GB cap.

Rule 3/4

No hardcoded row IDs, no manual test labels; every decision is a function of raw inputs plus declared external files (all cited in the paper).

Rule 6

Contested item (রক্তকমল: মধ্যপদলোপী vs উপমান কর্মধারয়) reported publicly on the Discussion tab.

Backup: known limitations

Poisoned gold-key entries

A small number of external QA entries are wrong and force incorrect labels on specific rows; the attempted fix (disagreement veto) failed its anchor gate, so the defect ships documented.

No-context weakness

Judge AUC is 0.748 on no-context rows — the binding constraint outside solver coverage.

Bengali-specific solvers

The routing principle transfers to other languages; the grammar solvers do not.

Small anchor

299 labeled rows can admit or kill a layer on noise — hence pre-committed criteria instead of repeated peeking.

Stated in the paper's Limitations section — consistent story across paper, slides, and Q&A.